Some required background

Linear algebra

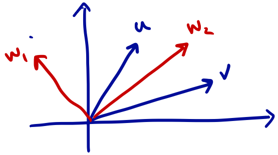
Basic multi-variable calculus

Basic understanding of algorithmic complexity

Linear Algebra Primer

Vectors in $\mathbb{R}^n$, $\mathbb{R}^d$ — $v \in \mathbb{R}^d$: $v = (v_1, \dots, v_d)$

linear (in)dependence, span, basis



$u, v \in \mathbb{R}^2$.

① $\{u, v\}$ span $\mathbb{R}^2$

$\forall w \in \mathbb{R}^2$, $\exists \alpha, \beta$ s.t.

$$w = \alpha u + \beta v$$

More generally: $\{v_1, \dots, v_k\}$ span $\mathbb{R}^k$ if $\forall w \in \mathbb{R}^k$, $\exists \alpha_1, \dots, \alpha_k$

$$w = \sum \alpha_i v_i$$

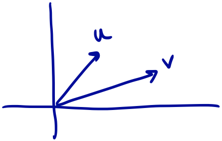
$\{v_1, \dots, v_k\}$ span V if . . .

Linear independence:

$v_1, \dots, v_k$ are linearly independent if there is no way to describe one of them as a linear comb. of the others.

In other words: If $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0$

then $\alpha_i = 0, \ \forall i.$



If $\alpha_1 u + \alpha_2 v = 0 \Rightarrow \alpha_1 u = -\alpha_2 v$

$\Rightarrow u = \left(-\frac{\alpha_2}{\alpha_1}\right) v$

Therefore, the only way for this equation: $\alpha_1 u + \alpha_2 v = 0$ to hold, is $\alpha_1 = \alpha_2 = 0$

$\Rightarrow u, v$ are independent.

Basis: If $v_1, \dots, v_n$ are independent & span $\mathbb{R}^n$, then they form a basis for $\mathbb{R}^n$.

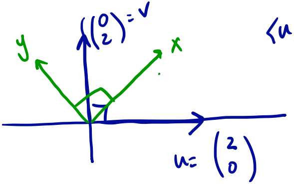
Linear Algebra Primer

$u, v \in \mathbb{R}^n$

dot / inner product : $u \cdot v = u^T v = \langle u, v \rangle = \sum_{i=1}^{n} u_i v_i$

outer product : $uv^T = \begin{pmatrix} | \\ u \\ | \end{pmatrix} \begin{pmatrix} — v — \end{pmatrix}$ n×n matrix

Cauchy-Schwarz : $\langle u, v \rangle \leq \|u\|_2 \cdot \|v\|_2$



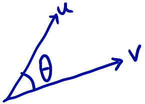
$\langle u, v \rangle = u_1 v_1 + u_2 v_2 = 0 \cdot 2 + 2 \cdot 0 = 0.$

$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \; y = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \quad x^T y = -1 + 1 = 0.$

Linear Algebra Primer

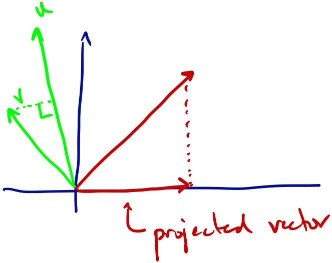
angle

u, v perp. $\iff \langle u, v \rangle = 0$



$$\cos \theta = \frac{\langle u, v \rangle}{\|u\|_2 \cdot \|v\|_2}$$

projection



Given vectors v, and u the projection of v onto u: $\dfrac{\langle v, u \rangle}{\|u\|_2^2} \cdot u$

Linear Algebra Primer

projection onto a subspace

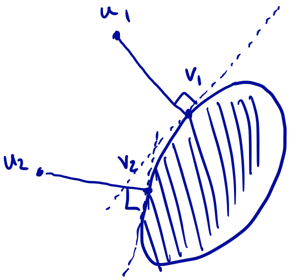
We can project onto a subspace spanned by vectors $u_1, \dots, v_k$

If $u_1, \dots, u_k$ are perpendicular to each other, and of length 1,

then
$$\mathrm{Proj}_{\{u_1, \dots, u_n\}}(v) = \langle v, u_1 \rangle \cdot u_1 + \langle v, u_2 \rangle \cdot u_2 + \dots + \langle v, u_k \rangle \cdot u_k$$

projection onto a convex set

$$\|u_1 - u_2\|_2 \geq \|v_1 - v_2\|_2$$

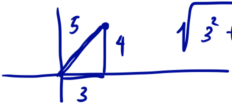


Linear Algebra Primer

$x \in \mathbb{R}^d$ $\quad \|x\|_2$ – "Euclidean Length" aka "Euc. norm"

L-p norms:

$$\|x\|_2 = \sqrt{\sum x_i^2}$$



$$\sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

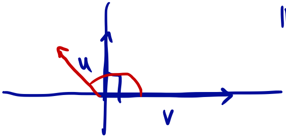
$$\langle x, x \rangle = \|x\|_2^2$$

For $1 \le p \le \infty$ $\quad \|x\|_p = \left(\sum |x_i|^p\right)^{1/p}$,

$\|x\|_\infty$ = largest (in abs. value) entry of x

$$= \max_i |x_i|$$

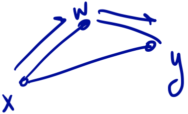
triangle inequality / Pythagorean theorem



$$\|u\|^2 + \|v\|^2 = \|u - v\|^2$$

If angle is $\ge 90^\circ$ $\Rightarrow$ $\|u\|^2 + \|v\|^2 \le \|u - v\|^2$

Triangle Ineq:

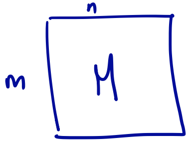


$$\|x - y\|_2 \le \|x - w\|_2 + \|w - y\|_2$$

Linear Algebra Primer

matrix rank, determinant, null space, range (1/2)

Matrix: $M \in \mathbb{R}^{m \times n}$



Null space (M): $\{x \in \mathbb{R}^n : Mx = 0\}$

Range (M): $\{y \in \mathbb{R}^m : y = Mx,$ for some $x \in \mathbb{R}^n\}$

Rank (M) = dimension of span of columns of M

= " " rows of M

If M is $n \times n$, $\text{rank}(M) = n$ $\underline{\underline{\text{iff}}}$ $\det M \neq 0$

iff $\text{null}(M) = \{0\}$

iff $\text{Range}(M) = \mathbb{R}^n$.

Linear Algebra Primer

matrix rank, determinant, null space, range (2/2)

If $\det M = 0$ (M is $n \times n$)

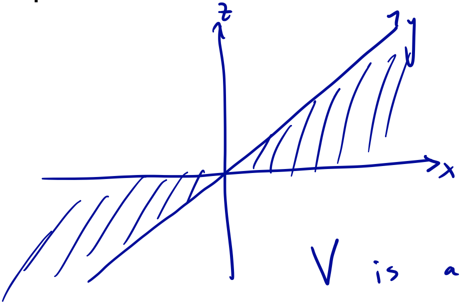
$\Rightarrow$ the columns of M cannot all be independent

also

the rows of M cannot all be independent.

Linear Algebra Primer

subspace



span $\{x, y\}$ is a subspace.

V is a subspace if it is closed under addition & scaling.

Linear Algebra Primer

orthogonal complement of a subspace

For V a subspace:

$$V^{\perp} = \{ w : \langle w, v \rangle = 0, \ \forall v \in V \}$$

affine subspace

subspace translated by a fixed vector.

Linear Algebra Primer

symmetric matrices, eigenvalues and eigenvectors (1/2)

M an $n \times n$ matrix, M is symmetric if $M = M^T$,

i.e., $M_{ij} = M_{ji}$.

(λ, v) is an eigenvalue-eigenvector pair for M if

$Mv = \lambda v$.

Ex: $M = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$v = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\lambda = 2$ is an eivector-eigenvalue pair. So is: $v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ $\lambda = 0$

symmetric matrices, eigenvalues and eigenvectors (2/2)

Fact: All eigenvectors of a symmetric matrix are perpendicular and all eigenvalues are real.

Ex: $M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ — check that e-values are <u>not</u> real (M not symmetric).

We can write: $\boxed{M = \sum \lambda_i v_i v_i^T}$, where (λ_i, v_i) is an eigenvalue-eigenvector pair for M, and $\|v_i\|_2 = 1$.

"Eigenvalue decomposition" / "spectral decomposition"

Linear Algebra Primer

M symmetric: $M = \sum \lambda_i v_i v_i^T$ outer product $n \times n$ matrix

positive semidefinite matrices (1/3)

If M is symmetric & all $\lambda_i \geq 0$ then

M is called <u>semidefinite</u>

positive semidefinite matrices (2/3)

A is psd iff the quadratic it defines is nonnegative $\circledast$

A is psd $\iff$ $A = \sum \lambda_i v_i v_i^T$, $\lambda_i \geq 0$

$\iff$ $x^T A x \geq 0 \quad \forall x \in \mathbb{R}^n$

Let's show this: Let's write x in terms of basis $\{v_1, \ldots, v_n\}$

$\Rightarrow \quad x = \sum \alpha_i v_i$ some α_i

$$x^T A x = \left(\sum \alpha_i v_i\right)^T \left(\sum_j \lambda_j v_j v_j^T\right)\left(\sum_k \alpha_k v_k\right)$$

$$= \left(\sum \alpha_i v_i\right)^T \cdot \sum_{j,k} \lambda_j \alpha_k v_j \underbrace{v_j^T v_k}_{=0 \text{ if } j \neq k,\ =1 \text{ if } j = k} = \left(\sum_i \alpha_i v_i\right)^T \sum_j \lambda_j \alpha_j v_j$$

$$= \sum_{i,j} \alpha_i \alpha_j \lambda_j \underbrace{v_i^T v_j} = \sum_i \underbrace{\lambda_i}_{\geq 0} \alpha_i^2 .$$

Linear Algebra Primer

positive semidefinite matrices (3/3)

for A any matrix, AA' and A'A are always psd

use previous fact:

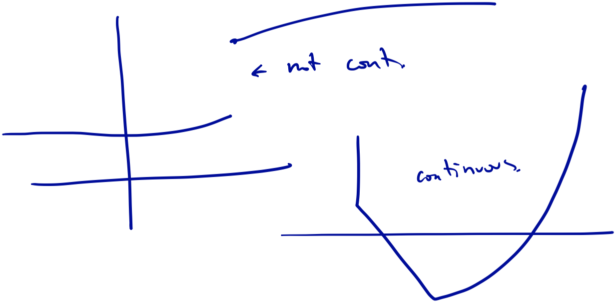
$$\longrightarrow x^T A A^T x = \|A^T x\|_2^2 \geq 0$$

" $v^T v$ "

similarly for $A^T A$.

Basic Multi-variable Calculus

Continuity :



Lipschitz continuity

f(x) is called is called Lipschitz continuous

if: $|f(x) - f(y)| \leq L \cdot \|x - y\|_2$

Basic Multi-variable Calculus

Gradient :

$$f : \mathbb{R}^n \to \mathbb{R} \qquad f(x) = f(x_1, \dots, x_n)$$

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

Hessian :

$$\nabla^2 f(x) = \begin{pmatrix} \\ & \frac{\partial^2 f}{\partial x_i \partial x_j} & \\ \\ \end{pmatrix}$$

Basic Multi-variable Calculus

Gradient/Hessian of a quadratic

$$f(x) = x^T Q x + q^T x + c$$

$$\nabla f(x) = 2Qx + q$$

$$\nabla^2 f(x) = 2Q$$

We will need other concepts...

...but we will develop them as needed, or provide some references...

In the meantime, some recommended references:

Introduction to Applied Linear Algebra, by Stephen Boyd and Lieven Vandenberghe
Convex Optimization, by Stephen Boyd and Lieven Vandenberghe
Linear Algebra Done Right, by Sheldon Axler